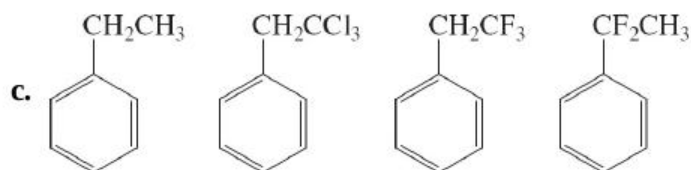
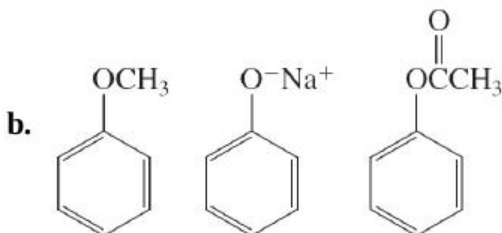
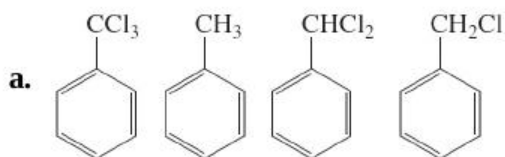
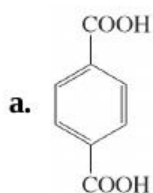


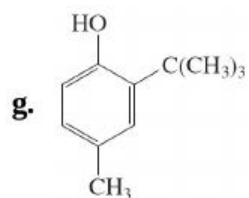
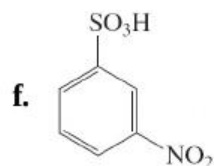
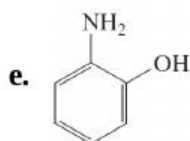
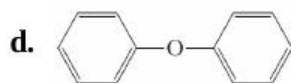
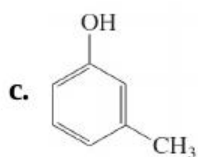
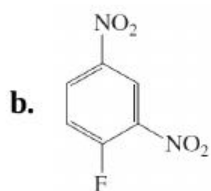
Problems

30. Rank the compounds in each of the following groups in order of decreasing reactivity toward electrophilic substitution. Explain your rankings.

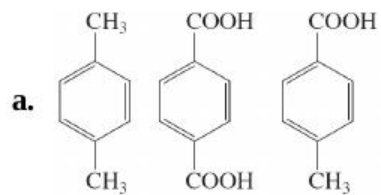


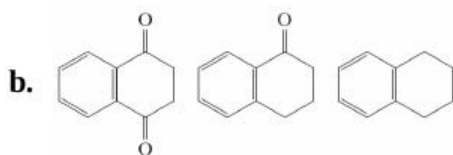
31. The rate of nitration of (chloromethyl)benzene, , is 0.71 relative to the rate of nitration of benzene (=1). The (chloromethyl)nitrobenzene product mixture that results contains 32% ortho, 15.5% meta, and 52.5% para isomers. Explain.
32. Specify whether you expect the benzene rings in the following molecules to be activated or deactivated.



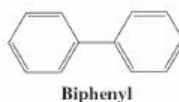


33. Rank the compounds in each of the following groups in order of decreasing reactivity toward electrophilic aromatic substitution. Explain your answers.



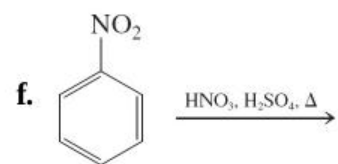
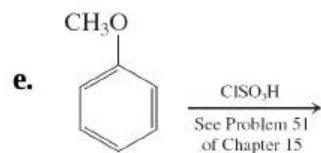
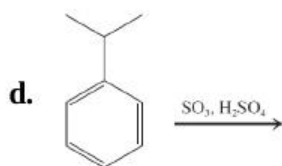
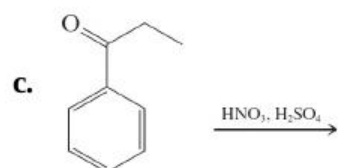
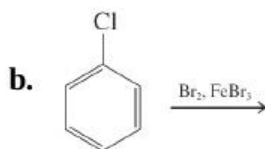
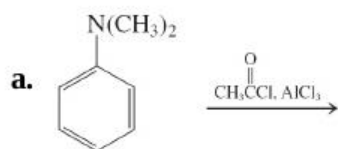


34. Halogenation of 1,3-dimethylbenzene (*m*-xylene) takes place 100 times faster than halogenation of either its 1,2- or 1,4-isomers (*o*- or *p*-xylene). Suggest an explanation.
35. Write the structure(s) of the major product(s) that you expect from each of the following electrophilic aromatic substitutions. **(a)** Nitration of methylbenzene (toluene); **(b)** sulfonation of methylbenzene (toluene); **(c)** nitration of 1,1-dimethylethylbenzene (*tert*-butylbenzene); **(d)** sulfonation of 1,1-dimethylethylbenzene (*tert*-butylbenzene).
- In what way does changing the substrate structures from methylbenzene (toluene) in (a) and (b) to 1,1-dimethylethylbenzene (*tert*-butylbenzene) in (c) and (d) affect the expected product distributions?
36. Write the structure(s) of the major product(s) that you expect from each of the following electrophilic aromatic substitutions. **(a)** Sulfonation of methoxybenzene (anisole); **(b)** bromination of nitrobenzene; **(c)** nitration of benzoic acid; **(d)** Friedel-Crafts acetylation of chlorobenzene.
37. Draw appropriate resonance forms to explain the deactivating meta-directing character of the SO_3H group in benzenesulfonic acid.
38. Do you agree with the following statement? “Strongly electron-withdrawing substituents on benzene rings are meta directing because they deactivate the meta positions less than they deactivate the ortho and para positions.” Explain your answer.
39. Draw appropriate resonance forms to explain the activating ortho, para-directing character of the phenyl substituent in biphenyl (below).



40. Give the expected major product(s) of each of the following

electrophilic substitution reactions.

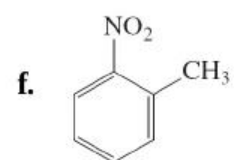
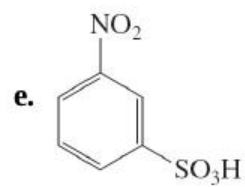
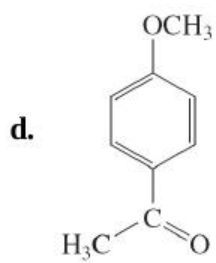
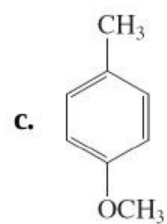
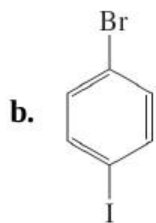
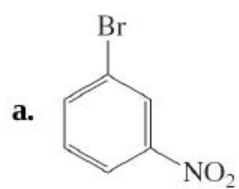


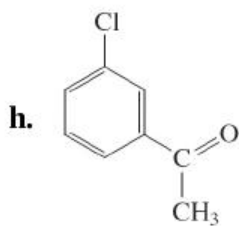
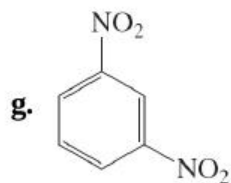
g. Write a detailed mechanism for each of the reactions in parts (a) through (f).

h. Propose a mechanism for the reaction of 1-methoxy-2-nitrobenzene with 2-propanol in the presence of HF, as illustrated in [Section 16-5](#).

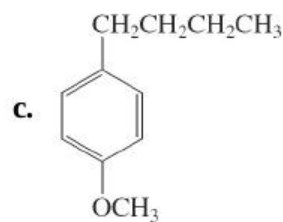
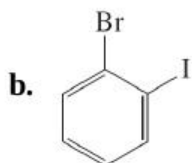
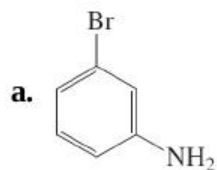
41. Reaction review. Without consulting the Reaction Road Map on [p. 749](#), suggest a combination of reagent and monosubstituted benzene that would give each of the following compounds. (**Hint:** Refer to [Table 16-2](#) for guidance. **Caution:** Directing effects arise from the group already

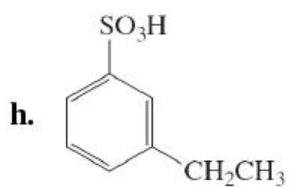
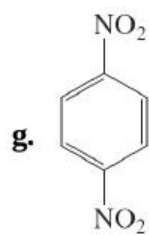
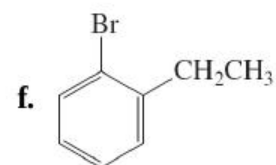
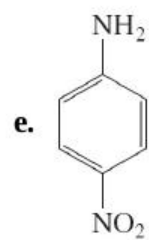
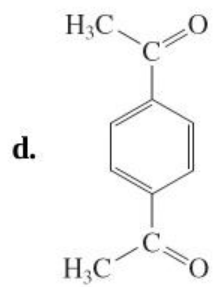
present on the benzene ring, *not* the group you are trying to attach.)



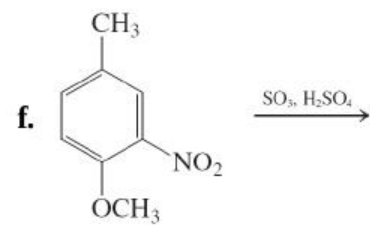
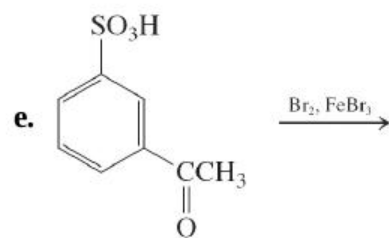
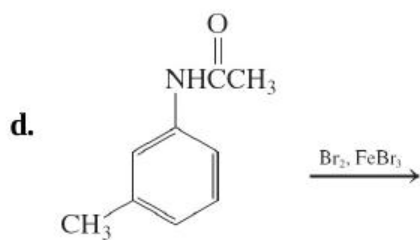
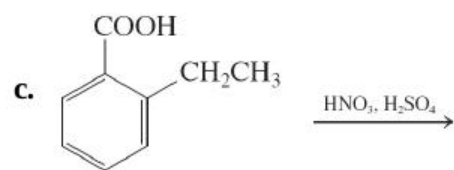
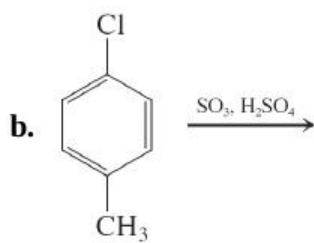
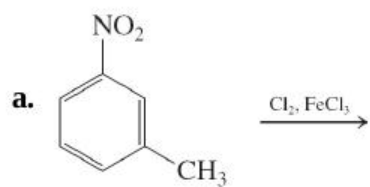


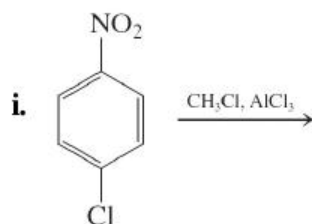
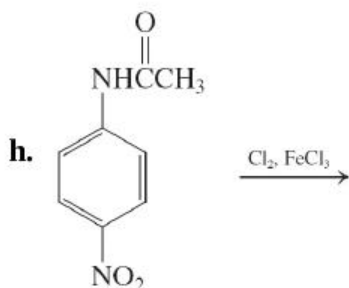
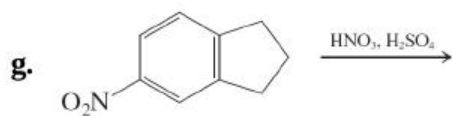
- 42.** Reaction review. The preparation of the following compounds requires more than one step. As in [Problem 41](#), suggest a monosubstituted benzene as starting material and the reagents for all the steps needed to complete their synthesis.



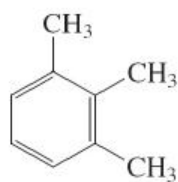


43. Give the expected major product(s) of each of the following reactions.

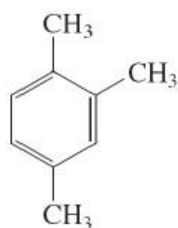




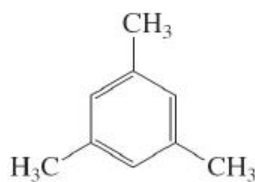
44. **CHALLENGE** (a) When a mixture containing one mole each of the three dimethylbenzenes (*o*-, *m*-, and *p*-xylene) is treated with one mole of chlorine in the presence of a Lewis acid catalyst, one of the three hydrocarbons is monochlorinated in 100%~~100%~~ yield, whereas the other two remain completely untouched. Which isomer reacts? Explain the differences in reactivity. (b) The same experiment carried out on a mixture of the following three trimethylbenzenes gives a similar outcome. Answer the questions posed in (a) for this mixture of isomers.



1,2,3-Trimethylbenzene

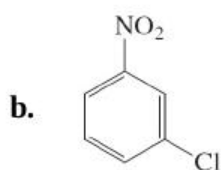
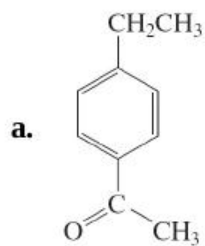


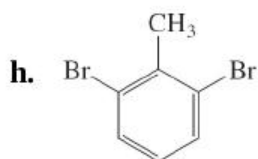
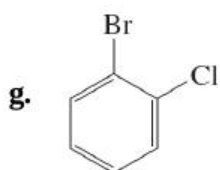
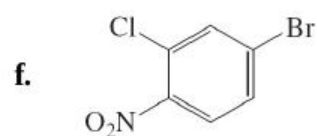
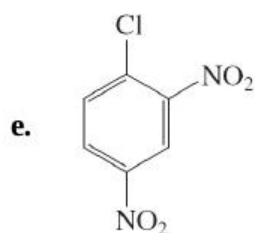
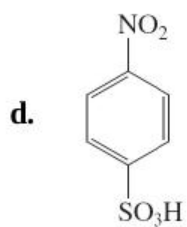
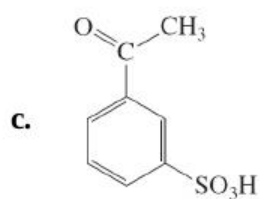
1,2,4-Trimethylbenzene



1,3,5-Trimethylbenzene

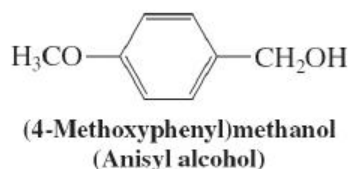
45. Propose a reasonable synthesis of each of the following multiply substituted arenes from benzene.





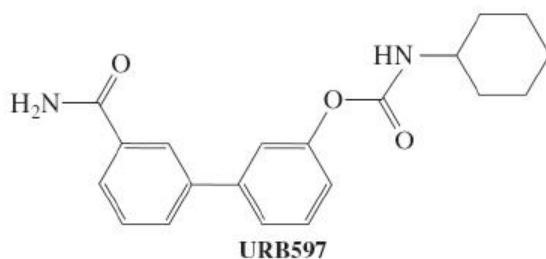
46. (4-Methoxyphenyl)methanol (anisyl alcohol) (see below) contributes both to the flavor of licorice and to the fragrance of lavender. Propose a synthesis of this compound from methoxybenzene (anisole). (**Hint:**

Consider your range of options for alcohol synthesis. If necessary, refer to [Problem 55](#) of [Chapter 15](#).)



47. Propose a one-step synthesis of sulfosalicylic acid ([p. 775](#)) from a suitable precursor.
48. The science of pain control has evolved dramatically in the past several years. The body deals with pain by releasing anandamide (see [Section 20-6](#)). Anandamide binds to the cannabinoid receptor, the same site that recognizes the active ingredients in marijuana. Binding to this site suppresses the perception of pain. The effect is not long lasting, because another enzyme degrades anandamide over time. Recent research has focused on finding therapeutic molecules capable of blocking this anandamide-destroying enzyme. The biphenyl derivative URB597 is an experimental substance that displays some of this desired enzyme-blocking ability, and it has been demonstrated to enhance pain suppression in rats.

Examine the structure of URB597 and address the following question: Would you expect URB597 to be an easy or a difficult target molecule to prepare, beginning with biphenyl ([Problem 39](#))? If you chose easy, illustrate how the reactions of this chapter might be employed in a synthetic approach to this drug. If you chose difficult, explain why.



49. The NMR and IR spectra for four unknown compounds A through D are presented below. Possible empirical formulas for them (not in any

particular order) are $\text{C}_6\text{H}_5\text{Br}$, $\text{C}_6\text{H}_6\text{BrN}$, $\text{C}_6\text{H}_5\text{Br}$, $\text{C}_6\text{H}_6\text{BrN}$, and $\text{C}_6\text{H}_5\text{Br}_2\text{N}$ (one of these formulas is used twice—two of the unknowns are isomers). In their ^{13}C NMR spectra, A, B, and D show four signals, and C displays six. Propose a structure and suggest a synthesis of each unknown, starting from benzene.

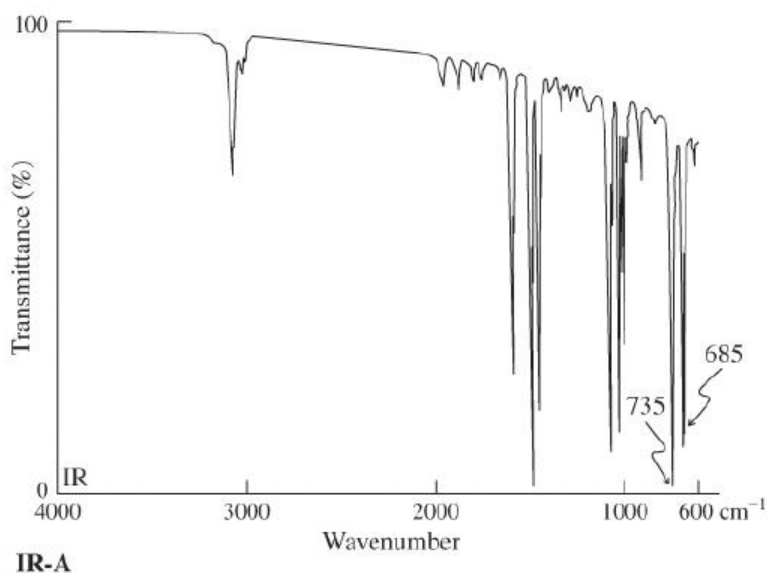
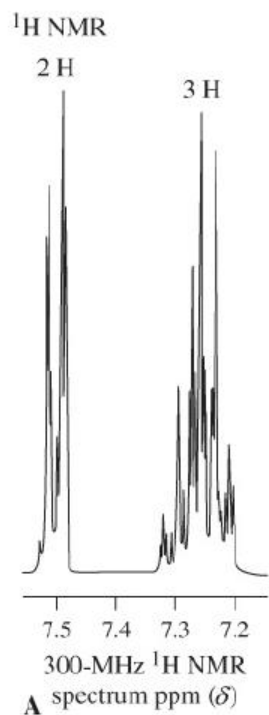
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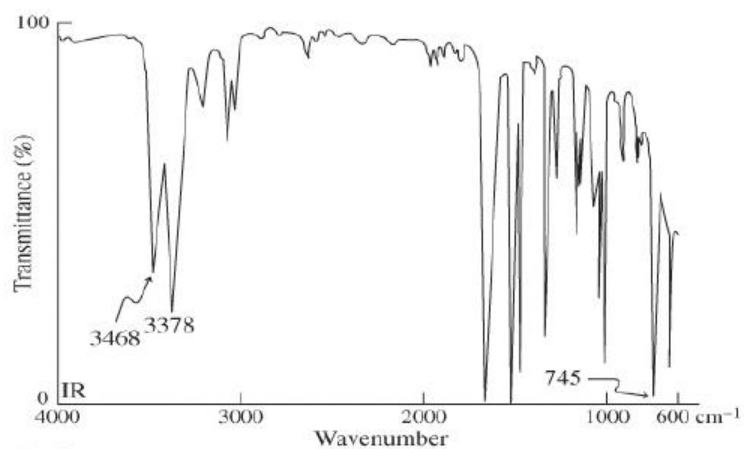
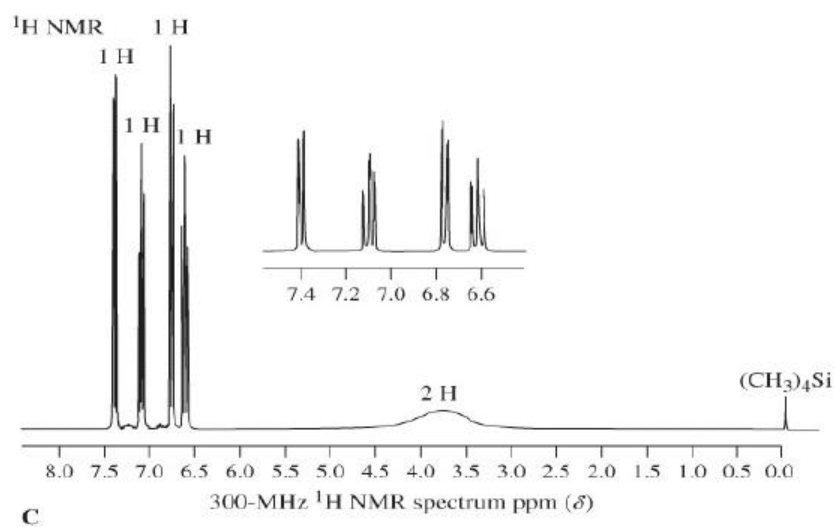
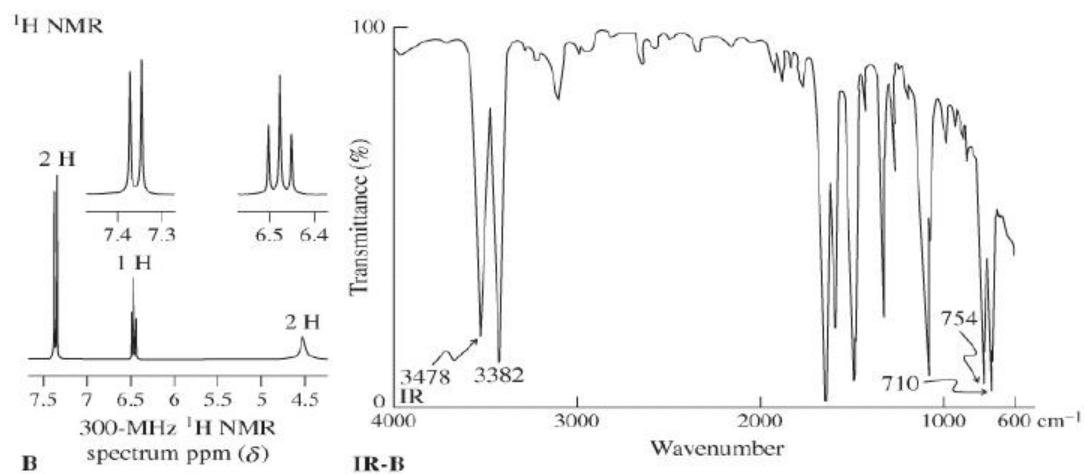
^1H NMR Information

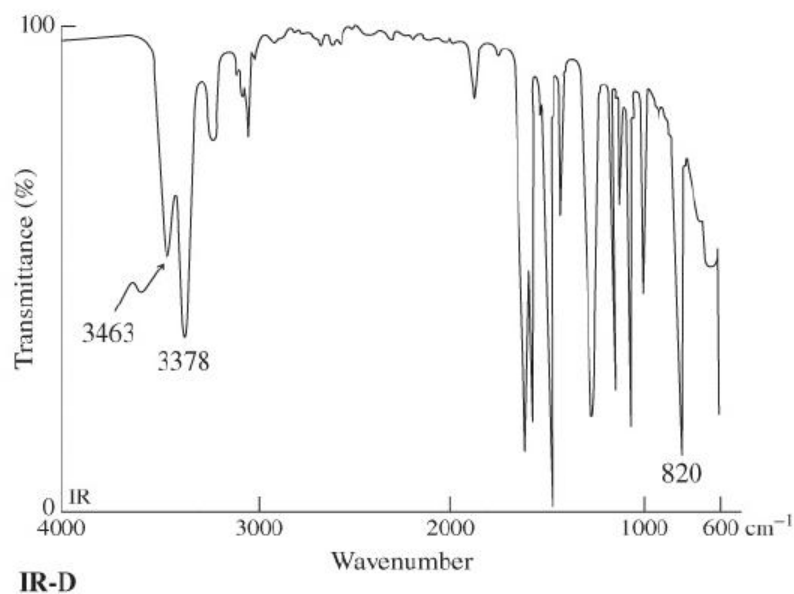
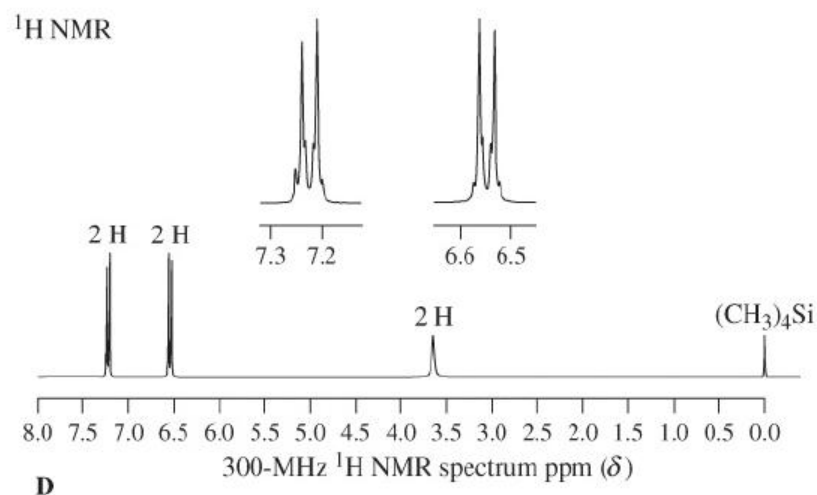
Chemical shift

Integration

Spin-spin splitting





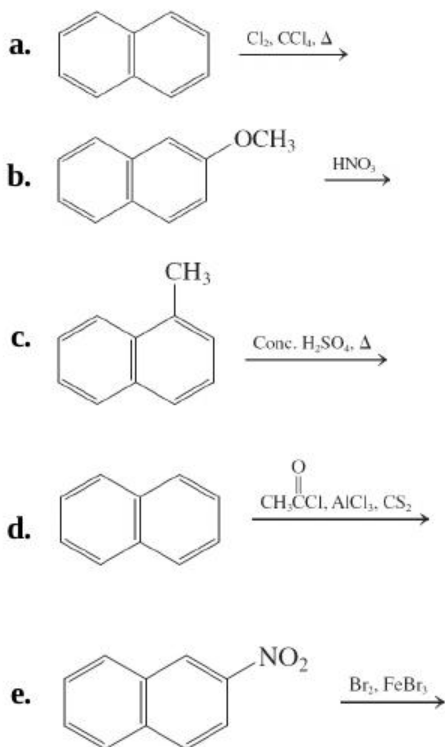


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50. Catalytic hydrogenation of naphthalene over Pd–CPd–C results in rapid addition of 2 moles of H₂. **H₂**. Propose a structure for this product.
51. Predict the major mononitration product of each of the following disubstituted naphthalenes. **(a)** 1,3-Dimethylnaphthalene; **(b)** 1-chloro-5-methoxynaphthalene; **(c)** 1,7-dinitronaphthalene; **(d)** 1,6-

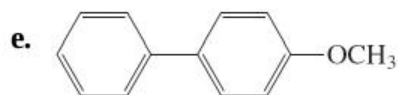
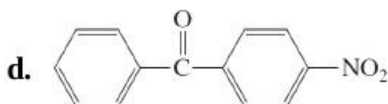
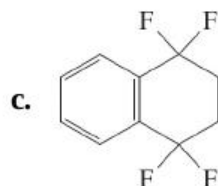
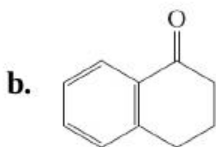
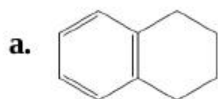
dichloronaphthalene.

52. Write the expected product(s) of each of the following reactions.



53. **CHALLENGE** Sulfonation of naphthalene at 80°C gives almost entirely 1-naphthalenesulfonic acid, whereas the same reaction at 160°C gives 2-naphthalenesulfonic acid. Propose an explanation. (**Hint:** For the underlying principle, see [Section 14-6](#).)
54. Electrophilic substitution on the benzene ring of benzenethiol (thiophenol, $\text{C}_6\text{H}_5\text{SH}$) is not possible. Why? What do you think happens when benzenethiol is allowed to react with an electrophile? (**Hint:** Review [Section 9-10](#).)
55. (a) Although methoxy is a strongly activating (and ortho, para-directing) group, the meta positions in methoxybenzene (anisole) are actually slightly deactivated toward electrophilic substitution relative to benzene. Explain. (b) Draw a set of potential-energy curves similar to those in [Section 16-2](#), comparing the electrophilic substitution of benzene with that of methoxybenzene (anisole) in ortho, para and meta positions.

56. Predict the result of mononitration of



57. **CHALLENGE** The *nitroso* group, $-\text{NO}$, $-\text{NO}$, as a substituent on a benzene ring acts as an ortho, para-directing group but is deactivating. Use the Lewis structure of the nitroso group and its inductive and resonance interactions with the benzene ring to explain this finding. (**Hint:** Consider possible similarities to another type of substituent that is ortho, para-directing but deactivating.)
58. Typical conditions for nitrosation are illustrated in the following equation. Propose a detailed mechanism for this reaction.